

OCIO LAYERED DMP — OCEAN CASTLE

The professional handoff: float in, float out, and one split that two layers can never drift across.

FILE `atlas_ocio_dmp_oceancastle_workflow.json` VERIFIED 2026-07-16

THE STORY

This is the color-managed pipeline: OCIORead loads the ACEScg EXR, AtlasRegisterPlate records the float original's path and colorspace, and AtlasAttachSourcePlate rides that registration on the solve — so the Nuke layer export references the ORIGINAL float plate with colorspace='ACEScg'. The 8-bit browser preview is never mistaken for final data.

The layering is the classic three-part DMP: a SAM3-segmented sky on its own card, the castle as foreground, the sea as background. The subtle engineering: ONE AtlasDepthBandSplit drives BOTH clean-plate layers via band_side (foreground/background). Because the split is an absolute metric distance — not a percentile over each layer's own pixel population — both layers resolve the identical boundary by construction. Band drift, the classic two-layer failure, is impossible rather than merely tuned away. Priorities run farthest-highest (sea 10 > castle 0 > sky -10), the seam doctrine that makes edge smears lose depth near-ties to the layer behind.

IN THE VIEWPORT — PROJECTED VS. THE GREY MESH



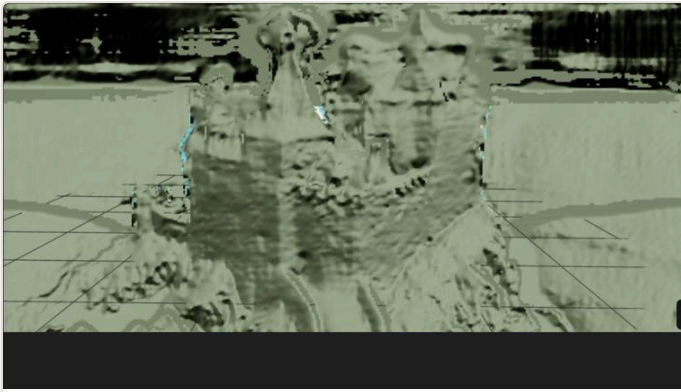
SOURCE PLATE

oceancastle_32bit_acescg.exr · ACEScg float — the single still photograph everything below is reconstructed from.



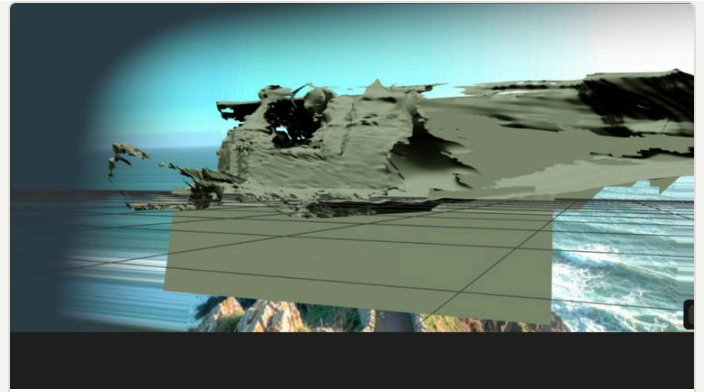
PROJECT ON

The photo reassembled on the recovered camera — texels assigned by ray, perfect from this view.



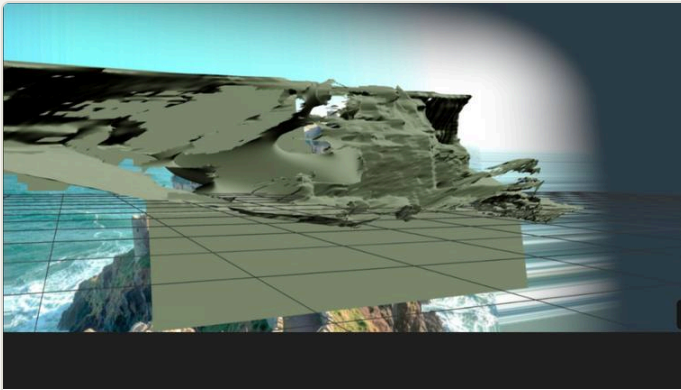
PROJECT OFF

The same geometry as a grey mesh, recovered view — what the depth-derived surfaces actually are.



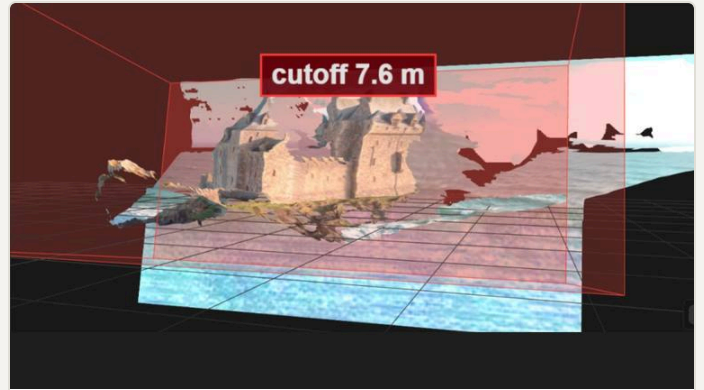
GREY · ORBIT LEFT

Orbited off-axis: the reconstructed cone. Coverage ends where the camera never saw.



GREY · ORBIT RIGHT

The opposite angle — silhouette tears read as deliberate holes, not errors.



BAND BOX

overlay: the translucent box's back face is the metric cutoff plane the layer is clipped at.

NODES ON STAGE

OCIORead

AtlasRegisterPlate

AtlasAttachSourcePlate

AtlasAssessImage

AtlasSolveGate

SAM3Segment

AtlasSkyDomeLayer

AtlasDepthBandSplit

AtlasCleanPlateLayer ×2

AtlasExportNukeLayers

AtlasExportUSD

RETUNE ON YOUR PLATE

- ▶ **EXR paths are absolute and per-machine.** Repoint OCIORead.source AND AtlasRegisterPlate.plate_path together at your copy.
- ▶ **The split percentile.** Where your foreground subject ends and background begins — the one number both layers share.
- ▶ **SAM3 sky prompt.** 'sky' works here; describe your plate's sky if it's unusual (nebula, cave opening...).

- **Don't invert the priorities.** Farthest-highest is load-bearing for seam rendering, not a style choice.
- **Two gates on first run.** The VLM's ► Continue (auto here) then ☒ Approve Solve — first Queue is a seconds-cheap preview.

RUN-VERIFIED

RUN ✓ 3 layers exactly as designed: sky 1.6k / castle 54k / sea 80k verts, farthest-highest. Per-layer plates + mattes + OBJs, the .nk, and camera.usda all written against the original EXR.

REQUIREMENTS

ComfyUI-OCIO + opencv 4.x + OPENCV_IO_ENABLE_OPENEXR=1 · ComfyUI-RMBG (SAM3) · LaMa · VLM optional · [usd]

Atlas Camera v0.6.0 beta · [examples/showcase](#)

[Master field guide →](#)